

BreatheBetter Spirometry Community Outreach

Prepared by:	Ryan Tyagi	Mentors:	Mr Ritesh & Ms Manu Kaundal
Programme:	BreatheBetter Spirometry Outreach	Focus Area:	Residential Society Lung Health

1. Programme Overview

The BreatheBetter Spirometry Community Outreach programme was designed and executed as an independent initiative to assess residential lung health. The core objective was to administer spirometry tests to community members within a local housing society. Using a self-built spirometer powered by an Arduino Uno and an HC-SR04 ultrasonic sensor, the programme collected real-world pulmonary function data to generate actionable community health insights.

2. Methodology

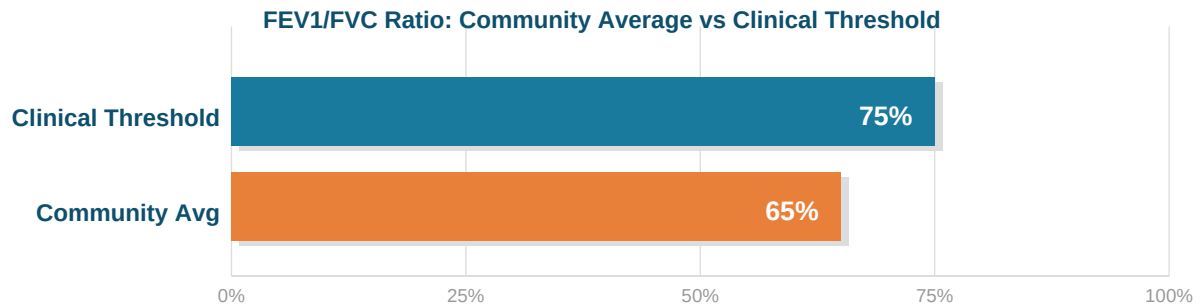
The spirometer device was constructed using an **Arduino Uno** microcontroller paired with an **HC-SR04 ultrasonic distance sensor**, which measured the displacement of a piston inside a calibrated tube to infer airflow volume. Two primary metrics were recorded for each participant:

Metric	Full Name	Clinical Significance
FEV1	Forced Expiratory Volume (1 sec)	Volume exhaled forcefully in one second; key indicator of airway obstruction.
FVC	Forced Vital Capacity	Total volume exhaled after maximum inhalation; measures lung capacity.
FEV1/FVC	Tiffeneau Index	Ratio below 75% may indicate obstructive lung disease; community average was 65%.

3. Findings and Analysis

Testing was conducted on over **15 residents** of the society, with a focus on older adults. The average FEV1/FVC ratio recorded across participants was **65%**. According to established clinical thresholds, a ratio below **75%** in adults is associated with reduced airway function and early signs of obstructive lung disease.

The findings suggest that a significant proportion of elderly residents in the tested community may be experiencing diminished lung function, rendering them physiologically vulnerable to respiratory illnesses. These results underscore the need for regular pulmonary screening and targeted health interventions within the residential community.



4. Device Build and Technical Challenges

The spirometer was independently designed and assembled using the following components:

Arduino Uno			Microcontroller — processes sensor readings and manages data output.
HC-SR04 Sensor	Ultrasonic		Measures piston displacement to calculate expiratory airflow volume.
Jumper Wires (x3 replaced)			Electrical connections between sensor, breadboard, and microcontroller.

Mid-campaign, a mechanical failure occurred — several jumper wires degraded and the sensor connection became unreliable, requiring a complete hardware refresh. Replacement parts were sourced and the device was reassembled, causing a brief operational pause before testing resumed. This experience reinforced the importance of hardware redundancy in field-deployed medical devices.

5. Skills Developed

Executing this project from concept to community deployment required the development of a broad and practical skill set:

- **Arduino Programming (C/C++):** Writing and debugging embedded code to capture and process ultrasonic sensor data in real time.
- **Electronic Circuit Assembly:** Building and troubleshooting breadboard circuits, managing sensor-to-microcontroller wiring.
- **Hardware Troubleshooting:** Diagnosing component failures under field conditions and performing mid-project repairs.
- **Medical Data Collection:** Administering standardised spirometry tests, recording FEV1 and FVC readings with participant consent.
- **Basic Data Analysis:** Calculating averages and interpreting FEV1/FVC ratios against clinical benchmarks.
- **Community Outreach and Communication:** Designing promotional flyers, disseminating information via society group messages, and liaising with the RWA.
- **Project Management:** Planning sessions, managing equipment logistics, and adapting the timeline following hardware failure.
- **Scientific Reporting:** Documenting methodology, findings, and conclusions in a structured, evidence-based format.

6. Outreach and Community Coordination

Community engagement was a central pillar of the programme. The following outreach activities were undertaken to maximise participation and ensure informed consent:

- Physical flyers were designed and displayed across common areas of the society, communicating the purpose, date, and voluntary nature of the testing.
- Digital announcements were shared via the official residential society WhatsApp group, ensuring broad and timely reach.
- Formal coordination was established with the Resident Welfare Association (RWA) to secure permissions, identify suitable testing venues, and facilitate participant recruitment.
- Mentors Mr Ritesh and Ms Manu Kaundal provided academic guidance and oversight throughout the planning and execution phases.

7. Recommended Next Steps

Based on the findings and experience gained, the following actions are recommended to deepen the impact of the BreatheBetter initiative:

- **Scale the testing cohort:** Expand participation beyond the initial 15+ residents to generate a statistically significant dataset.
- **Refer at-risk individuals:** Residents with FEV1/FVC ratios below 65% should be formally referred to a pulmonologist for clinical follow-up.
- **Device upgrade:** Solder all connections and house the device in a 3D-printed enclosure to prevent field failures.
- **Longitudinal tracking:** Re-test the same residents at six-month intervals to monitor changes in lung function over time.
- **Digital data dashboard:** Develop a simple visualisation interface to display anonymised, aggregated community lung health trends.
- **Educational workshop:** Host an awareness session at the society to share findings and provide guidance on breathing exercises and air quality.
- **Healthcare partnerships:** Collaborate with a local hospital or clinic for professional validation of device readings and follow-up care.
- **Project documentation:** Continue to document all iterations — hardware improvements, expanded data, community response — in structured reports.

The BreatheBetter Spirometry Community Outreach initiative successfully combined engineering, data collection, and community engagement to address a meaningful public health question. The discovery that the average FEV1/FVC ratio among tested elderly residents stands at 65% — below the 75% clinical threshold — represents a concrete, actionable finding with real implications for the community's health. Despite mid-project hardware setbacks, the programme was completed through perseverance, problem-solving, and the support of dedicated mentors. The skills and relationships developed provide a strong foundation for expanding this initiative into a sustained community health programme.





